Girish Kumar Patchikoru

AI/ML Engineer | Generative AI | Enterprise ML Systems

+1(361)282-7298 | [LinkedIn] | kumarpatchikoru7@gmail.com| Google Professional Cloud Architect | Azure AI Fundamentals

PROFESSIONAL SUMMARY

- **AI/ML Engineer** with **9+ years** of progressive experience across healthcare, insurance, banking, and financial services, evolving from classical **machine learning** and risk analytics into **MLOps** and production **Generative AI** systems.
- Hands-on experience building **Generative AI**, **Retrieval-Augmented Generation (RAG)**, and **agentic AI** workflows using **Python**, **LangChain**, and **LangGraph**, with **human-in-the-loop** controls for regulated healthcare and enterprise applications.
- Strong background in **predictive modeling** and applied **machine learning**, spanning **fraud detection**, **transaction-risk modeling**, **credit-risk assessment**, **identity resolution**, **anomaly detection**, **clustering**, and **underwriting analytics**.
- Applied **NLP** and **NLU** techniques across **clinical text processing**, **intent classification**, **fuzzy/phonetic identity matching**, and **conversational AI** use cases.
- Built **scalable ML, feature-engineering, and retrieval pipelines** using **Apache Spark**, **PySpark**, **Databricks**, **Python**, and **SQL** for structured and unstructured data supporting training, inference, and GenAI applications.
- Developed and integrated **production AI/ML services** through **FastAPI**, **RESTful APIs**, and **microservices**, supporting healthcare claims, eligibility, **prior authorization**, underwriting, and enterprise decision workflows.
- Worked across **AWS**, **Microsoft Azure**, **Google Cloud**, and **Databricks**, applying **MLflow**, **CI/CD**, **model monitoring**, **drift detection**, **Responsible AI**, **explainability**, and **human-in-the-loop governance** throughout the production ML lifecycle.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, PL/SQL, Pandas, NumPy, PySpark
- **Generative AI & LLMs:** Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, Prompt Engineering, Human-in-the-Loop (HITL), LLM Evaluation
- **Machine Learning & Deep Learning:** Scikit-learn, XGBoost, Logistic Regression, Random Forest, Isolation Forest, One-Class SVM, Classification, Anomaly Detection, Clustering, K-Means, SMOTE, Feature Engineering, Hyperparameter Tuning, Deep Learning
- **NLP & Text Analytics:** NLP, NLU, NLTK, SpaCy, TF-IDF, Intent Classification, Clinical Text Processing, Fuzzy/Phonetic Matching, Levenshtein Distance, Jaro-Winkler
- **MLOps & Model Lifecycle:** MLflow, Model Deployment, Model Serving, Model Validation, Model Versioning, Model Monitoring, Model/Data Drift Detection, Experiment Tracking, CI/CD, Automated Testing, Unit Testing, Integration Testing, Containerization, Production Support
- **Cloud & Data Platforms:** AWS, Microsoft Azure, Google Cloud, Databricks, Apache Spark
- **APIs & Integration:** FastAPI, RESTful APIs, Microservices, Model Inference Services, API Integration
- **Risk & Applied ML:** Fraud Detection, Transaction Risk Modeling, Credit Risk Modeling, Identity Resolution, Underwriting Analytics, Consumer Analytics, AML/KYC, Sanctions & PEP Screening, Rule-Based Engines (RBE), Claims & Eligibility Workflows, Prior Authorization
- **Responsible AI & Governance:** Responsible AI, Explainable AI, Model Governance, Bias Evaluation, Privacy, Model Documentation, Audit Support
- **Analytics & Visualization:** Statistical Analysis, Data Quality Validation, Tableau, Jupyter Notebook
- **Development & Collaboration:** Git, GitHub, Confluence

PROFESSIONAL EXPERIENCE

Senior AI-ML Engineer | United Health Care | Edina, Minnesota, US | Remote Dec 2025 – Present

- Architected and deployed **production Generative AI solutions** using **Python 3.12**, **Microsoft Azure**, and **Databricks**, supporting scalable healthcare operations and member-experience workflows.
- Built **Retrieval-Augmented Generation (RAG) applications** grounded in benefits, claims, provider, policy, and healthcare enterprise data, improving contextual relevance and traceability of LLM-generated responses.

- Developed **agentic AI workflows using LangChain 1.0 and LangGraph 1.0**, orchestrating retrieval, reasoning, tool execution, validation, and human escalation across complex healthcare administrative processes.
- Delivered **member-facing conversational AI capabilities** for benefits navigation, claims information, provider search, and cost guidance, enabling contextual handoff to customer advocates.
- Engineered **LLM evaluation workflows** for groundedness, retrieval relevance, response consistency, and hallucination risk across production GenAI applications.
- Integrated **Optum Real AI capabilities** into claims and reimbursement workflows, supporting earlier identification of prior-authorization, coverage, and payment issues.
- Designed **FastAPI 0.123-based REST APIs and integration services** for eligibility, benefits, claim-status, and prior-authorization workflows across payer, provider, and enterprise systems.
- Applied **NLP, NLU, and deep-learning techniques** to unstructured clinical text, supporting concept extraction, documentation analysis, and authorization-related decision support.
- Packaged **reusable AI services through the Optum AI Marketplace**, exposing standardized APIs and validating capabilities through sandbox and integration testing.
- Automated **digital prior-authorization workflows** using clinical, benefit, and policy data to identify missing documentation and coverage requirements before downstream review.
- Integrated **Google Cloud AI capabilities** into real-time payer operations, supporting automated eligibility, benefits, and claim-status workflows.
- Designed **scalable data and retrieval pipelines** using **Apache Spark/PySpark 3.5**, **Databricks**, **Python 3.12**, and **SQL**, preparing structured and unstructured healthcare data for training, inference, and GenAI retrieval.
- Built **production AI microservices using FastAPI 0.123**, incorporating **automated testing**, **containerization**, **Git/GitHub**, and **CI/CD** practices for controlled and repeatable releases.
- Implemented **ML and LLM observability frameworks**, tracking response quality, retrieval behavior, data quality, and inference latency across deployed AI services.
- Embedded **Responsible AI, privacy, security, explainability, and human-in-the-loop controls** throughout the AI lifecycle, aligning solutions with enterprise architecture and healthcare governance requirements.

Environment: Python 3.12, SQL, Generative AI, LLMs, RAG, LangChain 1.0, LangGraph 1.0, Apache Spark/PySpark 3.5, Databricks, Microsoft Azure, Google Cloud, FastAPI 0.123, REST APIs, Git/GitHub, CI/CD, Containerization, NLP, NLU, Prompt Engineering, LLM Evaluation, Model Monitoring, Responsible AI, Explainable AI.

AI - ML Engineer | Prudential Financial | Newark, NJ | Hybrid Jul 2024 – Nov 2025

- Operationalized **machine learning models for insurance underwriting and risk decisioning**, transitioning validated Data Science models into scalable production services.
- Developed **Python 3.11 and SQL inference components** for feature processing, model execution, and transformation of structured and semi-structured insurance datasets.
- Integrated **ML predictions with underwriting rules and decision workflows**, combining automated model scoring with human review for higher-risk cases.
- Built **distributed feature and data-processing pipelines** using **Apache Spark/PySpark 3.5** and **SQL**, preparing model-ready datasets for high-volume insurance workloads.
- Deployed and supported **production ML workloads on AWS**, maintaining scalable execution, environment consistency, and controlled model releases.
- Designed **RESTful model-serving APIs and microservices**, exposing underwriting predictions to enterprise applications and downstream decision systems.
- Developed **automated unit and integration testing frameworks** for feature transformations, model inference, API contracts, and downstream integrations.
- Implemented **CI/CD pipelines for ML applications**, automating validation and deployment controls across development, testing, and production environments.
- Managed **experiment tracking and model lifecycle workflows using MLflow 2.14**, supporting model comparison, versioning, reproducibility, and controlled promotion.
- Established **model and data-drift monitoring**, detecting changes in incoming features, prediction behavior, and production model performance.
- Instrumented **production inference services** for latency, service failures, and operational health, supporting root-cause analysis and production remediation.

- Prototyped **LangChain 0.3-based Generative AI workflow assistants**, evaluating response quality, reliability, and human-review requirements for enterprise automation use cases.
- Developed **model-governance and explainability artifacts**, documenting assumptions, validation results, known limitations, and observed production behavior to support governance and audit reviews.
- Evaluated **model performance and potential bias** across relevant populations, supporting **Responsible AI** reviews with Risk and Compliance stakeholders.

Environment: Python 3.11, SQL, Apache Spark/PySpark 3.5, AWS, MLflow 2.14, LangChain 0.3, REST APIs, Microservices, CI/CD, Automated Testing, Feature Engineering, Model Deployment, Model Validation, Model Versioning, Model Monitoring, Model/Data Drift, Explainable AI, Responsible AI, Git.

Data Scientist (ML) | TransUnion | Illinois, US Nov 2022 – Jun 2024

- Developed **machine learning models using Python 3.10 and Scikit-learn** for **credit-risk assessment** based on historical credit behavior, payment patterns, utilization, and account attributes.
- Engineered **predictive credit-risk features using Python 3.10 and SQL**, including payment history, credit utilization, account tenure, and delinquency indicators.
- Optimized **credit-risk classification models** through **feature selection and hyperparameter tuning**, evaluating performance using precision, recall, F1-score, and ROC-AUC.
- Analyzed **model stability across consumer segments**, reviewing score distributions and discrimination metrics to maintain reliable production decisioning.
- Incorporated **trended credit and alternative-data signals**, including payment behavior and ability-to-pay indicators, extending risk assessment for thin-file and new-to-credit consumers.
- Profiled and validated **large-scale consumer and credit datasets**, identifying predictive variables and resolving data-quality issues before model development.
- Translated **credit-risk model outputs into underwriting decision criteria**, connecting predictive scores with operational decisions alongside Risk and business stakeholders.
- Developed **identity-resolution workflows using matching and clustering techniques**, linking fragmented online and offline identifiers to individual consumers and households.
- Generated **identity-matching confidence scores** using clustering and similarity methods, supporting accurate entity linking and record deduplication.
- Validated **consumer identity attributes for TruAudience initiatives**, supporting identity matching, segmentation, and audience analytics.
- Supported **OneTru identity platform initiatives**, validating unified consumer attributes and model inputs across credit-risk and identity-resolution use cases.

Environment: Python 3.10, SQL, Scikit-learn 1.2, Machine Learning, Credit Risk Modeling, Identity Resolution, Classification, Clustering, Feature Engineering, Statistical Analysis, Model Validation, Hyperparameter Tuning, Model Governance, Explainable AI, Consumer Analytics

Data Scientist | JPMorgan Chase & Co. (JPMC)| Texas, US | Hybrid Nov 2020 – Oct 2022

- Developed and evaluated **fraud and transaction-risk ML models using Python 3.9 and Scikit-learn 1.0**, identifying suspicious customer behavior across large banking datasets.
- Engineered **transaction-level behavioral features** using **Python 3.9**, **Pandas**, **NumPy**, and **SQL**, including transaction velocity, account activity, and rolling behavioral aggregates.
- Built **customer-level risk indicators** from historical transaction data, capturing behavioral changes and longitudinal patterns for fraud-risk scoring.
- Trained and validated **classification and anomaly-detection models**, applying feature selection and hyperparameter tuning across model-development iterations.
- Evaluated **fraud-model performance** using precision, recall, F1-score, and ROC-AUC, balancing detection effectiveness against false-positive risk.
- Analyzed **account lifecycle and portfolio behavior** across underwriting, account management, and collections to identify fraud-risk drivers and changing customer patterns.
- Developed **scalable feature-engineering workflows** using **Apache Spark 3.2**, **Databricks**, and **SQL**, processing large structured banking datasets for model training and analytics.
- Partnered with Data Engineering on **legacy data and ML platform modernization**, migrating analytical datasets and ML workloads to AWS- and Databricks-based environments.

- Managed **experiment tracking and model reproducibility using MLflow**, capturing model parameters, performance metrics, and model versions across development cycles.

Environment: Python 3.9, SQL, Pandas, NumPy, Scikit-learn 1.0, Apache Spark 3.2, Databricks, AWS, MLflow 1.23, Tableau, Jupyter Notebook, Git.

Data Scientist | InfraSoftTech | Pune, India Apr 2017 – Sep 2020

- Developed fuzzy and phonetic identity-matching workflows using Python 3.6 for KYC screening against sanctions and Politically Exposed Person (PEP) watchlists within AML applications.
- Applied Levenshtein Distance and Jaro-Winkler similarity algorithms to customer-name matching, improving identity verification across inconsistent and partially matched records.
- Built anomaly-detection models using Scikit-learn 0.19, applying Isolation Forest and One-Class SVM to identify unusual transaction behavior beyond predefined screening rules.
- Enhanced Rule-Based Engine (RBE) transaction monitoring with statistical outlier-detection techniques, extending AML detection beyond deterministic rule thresholds.
- Developed fraud and transaction-risk classification models using Scikit-learn and XGBoost, applying Logistic Regression and Random Forest to historical banking transaction data.
- Applied SMOTE-based class balancing to highly imbalanced fraud datasets and engineered transactional features using Pandas and NumPy for rare-event detection.
- Built classical NLP workflows using NLTK 3.x, SpaCy 2.x, and TF-IDF, supporting text preprocessing, feature generation, and intent classification for banking chatbot use cases.

Environment: Python 3.6, Pandas, NumPy, Scikit-learn, XGBoost, Logistic Regression, Random Forest, Isolation Forest, One-Class SVM, SMOTE, K-Means, NLTK 3.x, SpaCy 2.x, TF-IDF, Levenshtein Distance, Jaro-Winkler, SQL, PL/SQL, Jupyter Notebook, Git, Confluence

CERTIFICATIONS:

- **Google Professional Cloud Architect Certification -** Google | Issued Feb 2026 · Expires Feb 2028
- **Prompt Engineering & Programming with OpenAI -** Columbia+ | Issued Sep 2025
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate -** Oracle | Issued Aug 2025
- **Oracle Cloud Infrastructure 2025 Certified Foundations Associate -** Oracle | Issued Aug 2025
- **Google Data Analytics Professional Certificate -** Google | Issued Jan 2025
- **AWS Certified Cloud Practitioner -** Amazon Web Services (AWS) | Issued Apr 2023
- **Microsoft Certified: Azure AI Fundamentals -** Microsoft | Issued Dec 2022
- **Microsoft Certified: Azure Data Fundamentals -** Microsoft | Issued Dec 2022
- **Microsoft Certified: Azure Fundamentals -** Microsoft | Issued Dec 2022

EDUCATION

- Bachelor of Technology (B. Tech) in Computer Science from KIET Group of Institutions, Uttar Pradesh, India